

STRESS INDUCED HEMATOLOGICAL CHANGES

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SUMMARY

Stress induced hematological changes such as clotting time, fibrinogen, thrombocyte, leukocyte, erythrocyte counts, hemoglobin concentration, glucose and corticosterone levels were observed in Sprague dwaley rats by forced swimming induced stress for 20minutes once and the same for two consecutive days.

In the case of applied stress for 20 minutes, decrease in clotting time and leucocyte count was observed with no change in hemoglobin concentration, whereas an increase in erythrocyte and thrombocyte counts, fibrinogen, glucose and corticosterone level was seen. After 20 minutes stress on each of the two consecutive days, the trend in change of all parameters remained the same except that there was an increase in hemoglobin concentration.

INTRODUCTION

Stress is a relative or transactional phenomena which indicates certain kinds of adaptive changes between an organism and its environment in response upon body to any burden either mental or somatic. Although stress is not a very specific term, it should not be mixed with anxiety which is some sort of uncertainty and helplessness. Fear and anxiety are the results of external and internal dangers respectively, whereas stress is a condition where body tends to adapt towards a change (Berger 1980). The hypothalamus can detect such adaptive change and initiates a chain of reactions which produces the general adaptation syndrome (Nakane et al., 1985).

The sympathetic adrenal system is activated when the environmental control is challenged, while the pituitary adrenal system is associated with the loss of control and the conservation withdrawal response. A variety of stressful events cause the release of a adrenocorticotrophic hormone (ACTH) from the anterior pituitary gland, which is regulated by a contricotropin releasing factor (CRF) form the hypothalamus (Nakane et al., 1985).

In acute stress corticosterone level increases, but on repeated stresses it lowers significantly as compared to acute stress (Armario et al., 1981) indicating adaptation

of hypothalamus-pituitary-adrenal system to repeated stresses. On repeated stresses, other changes also occur including the restoration of amine function (Herman et al., 1984). It has been suggested that the reduced transmitter levels due to acute stress or failure of adaptive neurochemical changes following chronic stress, favour the development or exacerbation of depressive symptoms (Zacharko et al., 1978).

In stress condition an increased levels of plasma free tryptophan and brain tryptophan, have also been reported (Dunn 1988 & Acworth et al., 1986). In stressful situations, a rapid release of CRF is noticed which has shown to stimulate the in-vivo GABA release (Sirinathsinghi 1988).

Stress can lead to certain diseases and disorders (Stein 1981) which might have profound impact on overall health through a variety of mechanisms (Weitzman et al., 1975), however the extent of stress, is a function of each individual's personality and physiological behaviour (Brophy 1990). Here we have tried to explore effects of stress in animals as models of human beings. We have evaluated some of the effects (changes in few hematological parameters) by swimming induced stress in rats. Glucose and corticosterone levels were determined, which act as an indicator of stress (Murison 1983).

MATERIALS

Chemicals of analytical grade of E. Merck or BDH, were used. Male Spargue Dwalery rats weighing 150-180 gms were used after keeping them in cages under a constant regulator light-dark cycles. Hemoglobin 3317 Merkotest kit, Hayems Fluid, and fluorometer were used for estimation of hemoglobin.

METHOD

Stress was induced by forced swimming method; subjecting a group of rats to swim in water, removing after 20 minutes, toweling and scarifying by decapitation. In the other set of rats, stress was induced in the same way but on two consecutive days. Effect of stress on various parameters such as clotting time, fibrinogen, erythrocyte and thromocyte counts, hemoglobin concentration, leukocyte, glucose and corticosterone levels were monitored consequent upon the applied stress in both the groups.

Blood clotting time was estimated by routine capillary method till the appearance of fibrin formation. Hemoglobin concentration was estimated by using a commercially available kit (3317 Hemoglobin Merckotest) and calculated by the following formula:

$$\text{Hb concentration} = \text{Absorbance} \times 36.8 \text{ g Hb/dl}$$

Leukocyte count was carried out by using Türk's solution and counting was carried out at about 100 times magnification. Blood erythrocyte count was carried out by using Hayem's fluid and then counting the cells in counting chamber. Thrombocyte count was also carried out similarly but in ammonium oxalate solution. Fibrinogen test was performed on citrated plasma, suspended in a water bath at 56°C for 15 minutes, then centrifuged at 2000 rpm for 10 minutes and measured the precipitated volume in ml. Glucose was estimated by O'toluidine method and corticosterone levels were determined by fluorometric method.

RESULTS AND DISCUSSION

The results obtained have been consolidated in Table 1. A significant decrease ($P < 0.005$) in clotting time on 20 minutes stress, and also the same in 20 minutes stress on the consecutive days was observed. No considerable change was observed in hemoglobin concentration after 20 minutes, but 20 minutes stress on 2 consecutive days showed a visible increase (< 0.01) in Hb concentration. The stresses produced a decrease ($P < 0.005$) in leukocyte count and a statistically significant increase ($P < 0.05$) in erythrocyte count was observed in both the cases. Both types of stresses caused increase ($P < 0.005$) in thrombocyte count, but no significant change was observed on fibrinogen contents after both of the stresses (Table 1). Glucose level was highly increased in both of the cases which is statistically important ($P < 0.005$) and the level of corticosterone was also increased ($P < 0.005$) during both types of stresses.

A significant suppression in clotting time was observed. In case of emotional stress, blood coagulation is accelerated, may be due to augmented secretion of epinephrine (Vosburah 1983), indicating some role of adrenal corticoid. Noradrenaline is increased during physical activity, whereas adrenalin during general emotional arousal. In anxiety, fear and danger also, the clotting time is shortened, the relative viscosity increased and blood pressure elevated. These effect might be a part of protective mechanism, and if become chronic, this pattern may lead to intravascular thrombosis.

An increase in red blood cell count after two days repeated stress causes the increased capillary filtration which may be the result of vasoconstriction (Jern 1989). Rise in hemoglobin concentration may also be due to the increase in PGE and a subsequent elevation of erythropoiesis. PGE₁ and PGE₂ activate erythropoiesis, hence the increased level of hemoglobin could be explained on the basis of increased PGE levels, after stress (Ikeda 1981). The study on the effect of hematocrit on cerebral flow (CBF) in man supports these findings. The elevated levels of corticoid suggest that corticoids like cortisol and corticosterone, give rise to an increased hemoglobin concentration and red blood cell count, which is also evident through frequent

occurrence of polycythemia in Cushing's syndrome (Bishop 1968). Corticoids have stimulating effect on erythropoiesis.

Corticoid also effect leukocyte count, because cortisol and corticosterone increase the number of polymorphonuclear leukocytes, as a result of the enhancement in the rate of their entrance into the blood streams from the bone marrow and relatively slow removal from the circulation (Bishop et al., 1968).

It has been revealed that assurance of coronary heart diseases and angina pectoris always enhance in stress. Acute psychological and mental stress may also have a direct bearing on coronary occlusion. During the present study increased number of platelets has also been found. The clotting system is activated by factor III released from platelets and thrombin converted to fibrin leads to further aggregation of platelets (Walter et al., 1987), which is then related to coagulation time.

A significant decrease in number of leukocytes was noted in stress condition, which is not in agreement with previous reports (Friedman et al., 1974). The immediate lymphocytosis is possible due to direct effect of adrenalin on leukocyte migration whereas second phase may be due to cortisol mediated from the bone marrow. Leukocytosis during stress stimulus seems to be neurogenic in origin and the present study has shown a positive correlation of hematological changes during stress which is in accordance to earlier findings (Jern et al., 1989).

Catecholamine and glucocorticoids both are released in response to stress and have a role in regulation of glucose metabolism (Jafferys & Funder, 1991). Glucocorticoids and stress hormones promote gluconeogenesis by inducing the synthesis of enzymes involved in gluconeogenesis (Baxter & Tyrell, 1987). In the present study, increased levels of glucose were observed, because in stress, hormone also increases glycogen in plasma and glycogen itself stimulates gluconeogenesis.

There have been a number of studies correlating cortisol production and psychiatric state. Emotional arousal has shown upto 25% elevation in cortisol production (Sacher, 1970). Increase corticosterone level observed by us, is in accordance with the previous studies (Amiragova 1979). It has been reported that corticosterone and prolactin elevations occur in response to stress stimuli (Brown & Martin 1974). The *in vitro* corticosterone production is a reliable index of hypothalamus-pituitary adrenal activity, while it has been suggested that corticosterone can be considered as an index of stress only in acute situations (Murison 1983).

In conclusion, it can be said that stress associated biological changes can lead to

many uncontrolled occupational disorders including coronary heart disease and mental problem etc. The present study when seen in conjugation with the earlier findings thus suggests, a timely, accurate and critical evaluation and check of the stress induced parameters which otherwise if ignored can lead to serious consequences.

Table 1
Effect of stress on different hematological parameters

Parameters	Control	20 minutes stress	20 minutes stress on two consecutive days
Clotting Time	2.5 Sec. (0.35)	1.4 Sec (0.1)	1.43 Sec (0.083)
Fibrinogen	0.316 ml (0.037)	0.312 ml (0.042)	0.322 ml (0.043)
Thrombocyte Count	2.512 mill/cu.mm (0.49)	5.78 mill/cu.mm (0.55)	6.54 mill/cu.mm (0.515)
Leucocyte Count	12,222 th/cu.mm (4.92)	7138 th/cu.m (4.32)	6626th/cu.m (5.94)
Erythrocyte Count	5.892 mill/cu.mm (0.44)	8.53 mill/cu.mm (0.382)	7.51 mill/cu.mm (0.30)
Hemoglobin Conc.	15.892 g/dl (1.74)	16.22 g/dl (3.97)	18.32 g/dl (0.63)
Glucose Levels	79.4 mg/dl (1.14)	156.6 mg/dl (7.0)	162 mg/dl (2.91)
Corticosterone Level	20.7 ug/ml (5.6)	56.6 ug/ml (7.0)	61.2 ug/ml (6.9)

*Each value is the mean of 5 independent experiments.

*Within the parentheses are the values of standard deviation.

REFERENCES

- Acworth I., Nicholan J., Morgan B. and Newsholme E.A. (1986). *Biochem. Biophys. Res. Commun.* 137(1): 149.
- Amiragova M.G. (1979). *Sechenova*. 65(5): 594.
- Armario A., Catellanos J.M. and Balasch J. (1981). *Horm. Metab. Res.* 13(7): 413.
- Baxter J.D. and Tyrell J.B. (1987). The adrenal cortex in endocrinology in metabolism. Felia P., Baxter J.D., Broadus L.A. and Froham L.A. (eds.) McGReshill, USA. p.287.
- Berger F.M. (1980). *Behav. Sci.* 25(4): 315.
- Bishop C.R., Athens J.W., Boggs D.R., Wanner H.R., Cartwright G.E. and Wintrobe M.M. (1968). *J. Clin. Invest.* 47: 249.
- Brophy J.T. (1990). Psychiatric disorder in current medical diagnosis and treatment. Appleton and Lang, New York, USA.
- Brown G.M. and Martin J.B. (1974). *Psychosom. Med.* 36(3): 241.
- Dunn A.J. (1988). *Life Sci.* 42(9): 1847.
- Friedman G.D., Klatsky A.L. and Seigelamb A.B. (1974). *J. Med.* 290: 1275.
- Herman J.P., Stimus L. and Le Moal M. (1984). *Psychopharmacol.* 84: 431.
- Ikeda Y.A. (1981). *Tokyo Med. Coll.* 39(2): 287.
- Jern C., Wadenvik M.H., Hallgren J. and Jein S. (1989). *Br. J. Haematol.* 71: 153.
- Jafferys D. and Funder J. (1991). *Eur. J. Pharm.* 205: 267.
- Murison C.R. (1983). *Psychology and Psychiatry.* 11: 20.
- Nakane T., Andhva T. and Kanie N.H.C. (1985). *Proc. Natl. Acad. Sci. USA* 82(4): 1247.
- Sacher E.J. (1970). *Brain Res.* 32: 316.
- Sirinathsinghi D.J.S. and Heaes R.P. (1988). *Neuroscience.* 1: 203.
- Stein M., Keller S. and Schleifer S. (1981). The hypothalamus and the immune response. Weinen. H., Hofrer M. and Stunkard A. (eds.) Brain, behaviour and bodily disease. New York, Raven, USA.
- Vosburah C.H. and Richard A.N. (1983). *Am. J. Physiol.* 9: 35.
- Walter J.B. and Israel M.S. (1987). General Pathology (6th edition). Edinburgh Churchill Livingstone, London.
- Weitzman E.D., Boyar R.M., Kapen S. and Hellman L. (1975). *Recent Progress Hormone Research.* 31: 399.
- Zacharko R.M. and Anisman H. (1978). Pharmacological, biochemical and behavioral analysis of depression in animal models. In animal models of depression. Kools G., Ellers C. and Bloom F.E. (eds). Boston Birkhauser. USA, p.121.